

Appendix: The Bias– Variance Decomposition

INFO 521 · Module 2 · Lecture B — supplement

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Where we are

Lecture B stated that, at a fixed input $\mathbf{x}$, expected test error splits into three pieces:

$$\mathbb{E}[(y - \hat{f}(\mathbf{x}))^2] = \text{bias}^2 + \text{variance} + \sigma^2.$$

This appendix derives that in full. Bookkeeping first — at the fixed $\mathbf{x}$:

- **Truth:** $y = f(\mathbf{x}) + \varepsilon$, with $\mathbb{E}[\varepsilon] = 0$, $\mathbb{E}[\varepsilon^2] = \sigma^2$.
- **Estimate:** $\hat{f} = \hat{f}_{\mathcal{D}}(\mathbf{x})$ — random *through the training set* $\mathcal{D}$ (m3b's repeated-datasets picture, in prediction space).
- **Average fit:** $\bar{f} = \mathbb{E}_{\mathcal{D}}[\hat{f}]$ — the orange curve from Lecture B's repeated-fits figure.
- ε is a **new** test-time noise draw, independent of $\mathcal{D}$.

The one trick: add and subtract $\bar{f}$

Write the miss as three centered pieces:

$$y - \hat{f} = \underbrace{(f - \bar{f})}_{\text{systematic miss}} + \underbrace{(\bar{f} - \hat{f})}_{\text{dataset luck}} + \underbrace{\varepsilon}_{\text{fresh noise}}.$$

Square and take the expectation (over $\mathcal{D}$ and ε):

$$\mathbb{E}[(y - \hat{f})^2] = \mathbb{E}[(f - \bar{f})^2] + \mathbb{E}[(\bar{f} - \hat{f})^2] + \mathbb{E}[\varepsilon^2] + (\text{three cross terms}).$$

The cross terms die

Take each in turn:

$$\mathbb{E}[(f - \bar{f})(\bar{f} - \hat{f})] = (f - \bar{f}) \mathbb{E}_{\mathcal{D}}[\bar{f} - \hat{f}] = (f - \bar{f}) \cdot 0 = 0$$

— $f - \bar{f}$ is a **constant** (no randomness left), and $\hat{f}$ is centered on $\bar{f}$ *by definition*.

$$\mathbb{E}[(f - \bar{f})\varepsilon] = (f - \bar{f}) \mathbb{E}[\varepsilon] = 0, \quad \mathbb{E}[(\bar{f} - \hat{f})\varepsilon] = \mathbb{E}_{\mathcal{D}}[\bar{f} - \hat{f}] \mathbb{E}[\varepsilon] = 0$$

— the test-time noise is mean-zero and **independent of the training set**, so both ε cross terms factor and vanish.

The decomposition, earned

What survives:

$$\mathbb{E}[(y - \hat{f}(\mathbf{x}))^2] = \underbrace{(f(\mathbf{x}) - \bar{f}(\mathbf{x}))^2}_{\text{bias}^2} + \underbrace{\mathbb{E}_{\mathcal{D}}[(\hat{f}(\mathbf{x}) - \bar{f}(\mathbf{x}))^2]}_{\text{variance}} + \underbrace{\sigma^2}_{\text{irreducible}}$$

- **bias²** — how far the *average* fit sits from the truth: rigidity's price.
- **variance** — how much a single fit swings around that average: flexibility's price.
- σ^2 — the noise floor; no model, however tuned, goes below it.

Complexity moves the first two in opposite directions; their sum is the U-curve of Lectures A/B.
Ridge's λ walks you along it.

Fine print

Three things the derivation quietly fixed — say them out loud once:

- **One $\mathbf{x}$ at a time.** The decomposition holds pointwise; “test error” in practice averages it over the input distribution.
- **Squared loss only.** The clean three-way split is a property of the square; other losses decompose differently (or not at all).
- **Expectations, not your dataset.** Bias and variance describe the *procedure* across repeated datasets — on your one sample you see a single draw, which is why CV (m2a) estimates rather than computes them.